



OSMP Implementation Description Guide

NAVAIR's Approach to MOSA using an
Open System Management Plan (OSMP)



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Rich Ernst	NAVAIR	5/10/2025	Baselined document for v.16 review
Dr. Steven Davidson	DASN/MITR	5/26/2025	Updated multiple section comments
Rich Ernst	NAVAIR	5/30/2025	Final review v.18
Rich Ernst	NAVAIR	6/03/2025	Final formatting v .2
Leadership	NAVAIR/NAWC	TBD	Signature v 1.0

Foreword:

As we navigate the complex landscape of modern warfare, the Navy and its aviation forces remain crucial to securing our nation's defense. In this era of rapid technological advancement, it is vital that we equip our warfighters with the best tools available while ensuring responsible management of taxpayer dollars. We are at a defining moment, where the choices we make today will shape the effectiveness of our military for years to come.

At the heart of this transformation is the Modular Open Systems Approach (MOSA). MOSA is more than just a design framework and acquisition approach—it is a strategic commitment to the future. By embracing open architectures and modular systems, we enhance flexibility, interoperability, and innovation, allowing for the seamless and rapid integration of new technologies without costly overhauls. This ability to adapt quickly to technological advancements is essential in maintaining a competitive edge in an ever-evolving defense environment.

Given the constraints on our defense budgets, managing costs while continuously innovating is crucial. The Open Systems Management Plan (OSMP) serves as the blueprint for this journey, aligning our acquisition strategies with goals of cost optimization, increased competition, and long-term sustainability. Through its structured framework, OSMP facilitates the integration of modular solutions, enabling us to enhance system capabilities affordably, while avoiding dependence on proprietary technologies.



This transformation requires a strong partnership between the Navy, industry partners, and the warfighter. By adopting MOSA, we provide our warfighters with adaptable, cutting-edge systems that can evolve with their needs and the challenges they face. The warfighter deserves tools that not only perform today but are ready for tomorrow's mission.

As we move forward, we must continually innovate, adapt, and ensure that every decision made reflects our commitment to both the warfighter and the efficient use of taxpayer resources. By embracing these principles of flexibility, innovation, and fiscal responsibility, we ensure the future success of Navy aviation and the readiness of our warfighters to meet the challenges ahead. This is the foundation for securing our nation's defense and values in the 21st century.

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Acronym list

AMS-GRA: Agile Mission Suite – Government Reference Architecture
ASN(RDA): Assistant Secretary of the Navy for Research, Development, and Acquisition
BDFD: Business-Driven Functional Decomposition
CA: Contract Award
CDRL: Contract Data Requirements List
COTS: Commercial Off-The-Shelf
CSCI: Computer Software Configuration Item
CTR: Contractor
DASN: Deputy Assistant Secretary of the Navy
DID: Data Item Description
DoD: Department of Defense
FACE: Future Airborne Capability Environment
GAO: Government Accountability Office
ICD: Interface Control Document
IP: Intellectual Property
MOSA: Modular Open Systems Approach
MOSAACQ: Modular Open Systems Approach to Acquisition
MSI: Modular Systems Interface
NAVAIR: Naval Air Systems Command
NIST: National Institute of Standards and Technology
OA: Open Architecture
OMS: Open Mission Systems
OSMP: Open Systems Management Plan
OSMPA: Open Systems Management Plan Data Item Description
OSD: Office of the Secretary of Defense
OSA: Open Systems Architecture
POSIX: Portable Operating System Interface
RFP: Request for Proposal
SME: Subject Matter Expert
SOSA: Sensor Open Systems Architecture
SOW: Statement of Work
TIM: Technical Interchange Meeting
TRF: Technical Reference Framework
UCI: Universal Command and Control Interface
U.S.C.: United States Code
VVT: Verification and Validation Testing
WOSA: Weapon Open Systems Architecture
XML: Extensible Markup Language

Introduction

The OSMP lays out a clear and straightforward process that helps programs take a consistent and effective approach to fine-tune the acquisition strategy in alignment with MOSA principles and objectives. It encourages program and technical leaders to ask the right questions and think strategically about how to shape their MOSA approach, enabling them to circumvent potential issues early and proactively. This guide outlines best practices for leveraging the OSMP effectively, addressing critical considerations including:

- The role of MOSA in fostering interoperability and cost efficiencies by integrating open standards across the Department of Defense (DoD) and commercial enterprises.
- Frequent errors in MOSA implementation and strategies to avoid them.
- Enhancements in the latest OSMP Data Item Description (DID) [DI-MGMT-82099 Rev. A OSMP].

The increasing significance of MOSA in acquisitions is underscored by evolving DoD policies and service-level guidance, solidifying the OSMP as a crucial contractual requirement and program management tool. Official directives, such as the December 2024 Navy MOSA Guidebook by the Assistant Secretary of the Navy for Research, Development, and Acquisition (ASN(RDA)), emphasize the necessity of the OSMP in contract deliverables, ensuring vendors uphold MOSA principles throughout the program lifecycle.

The OSMP serves as a critical program management tool and contractual deliverable, facilitating the promotion and oversight of openness and modularity decisions. To maximize its benefit, the government develops the initial OSMP draft, covering key government business and technology goals. This draft will be iteratively updated, starting with a targeted version at the proposal stage to support source selection evaluation. Subsequent to source selection, the contractor delivers updated versions at key milestones, which are subject to government concurrence. The OSMP is meant to be a collaborative "living" document between the government and contractor throughout the program lifecycle.



This document was developed to support the Naval Air Systems Command (NAVAIR) community in growing its MOSA approach across its portfolio of weapon systems. The following outlines the flow of each section:

1. **Introduction to NAVAIR's MOSA Process Framework:** This section introduces the concept of MOSA and its importance, outlining the five key phases of the framework that are instrumental in driving execution.
2. **Explanation of MOSA and Linking the MOSA Implementation Process to the OSMP:** This section provides the basics of MOSA and how the Open Systems Management Plan (OSMP) supports MOSA development and integration in a program.

3. **What is the Open System Management Plan (OSMP):** This section defines the OSMP, its purpose, and its key components.
4. **NAVAIR's OSMP Process Framework:** This section outlines the framework for developing an OSMP document, ensuring alignment with business goals and MOSA principles.
5. **Strategic Alignment of the OSMP and DoD Acquisition 5000 Framework:** This section emphasizes the importance of aligning the OSMP with the DoD 5000 framework from the earliest stages of acquisition.
6. **Proposal Evaluation and Contract Award Guide:** This section provides guidance on evaluating proposals based on their open systems strategies, including recommended RFP language.
7. **Key Business and Technical Drivers to Consider in an OSMP:** This section highlights the key business and technical considerations for a successful OSMP, emphasizing business-driven functional decomposition and verification & validation testing.
8. **Final Thoughts:** This section summarizes the key takeaways and emphasizes the critical role of the OSMP in achieving MOSA objectives.

The OSMP provides a well-defined concise process that leads to each program taking a consistent effective approach to optimize the acquisition strategy. It leads programs to ask the right questions to think strategically about structuring their acquisition strategy in a way that allows them to identify common pitfalls early and provides the tools to navigate mitigating them.



NAVAIR’s Modular Open Systems Approach (MOSA) Process Framework

Implementing a Modular Open Systems Approach (MOSA) requires adherence to DoD directives while adopting an Open Business Approach. A structured methodology ensures that MOSA objectives align with acquisition strategies and program execution. The following framework outlines (Figure 1) the sequential steps necessary to establish a robust MOSA strategy.

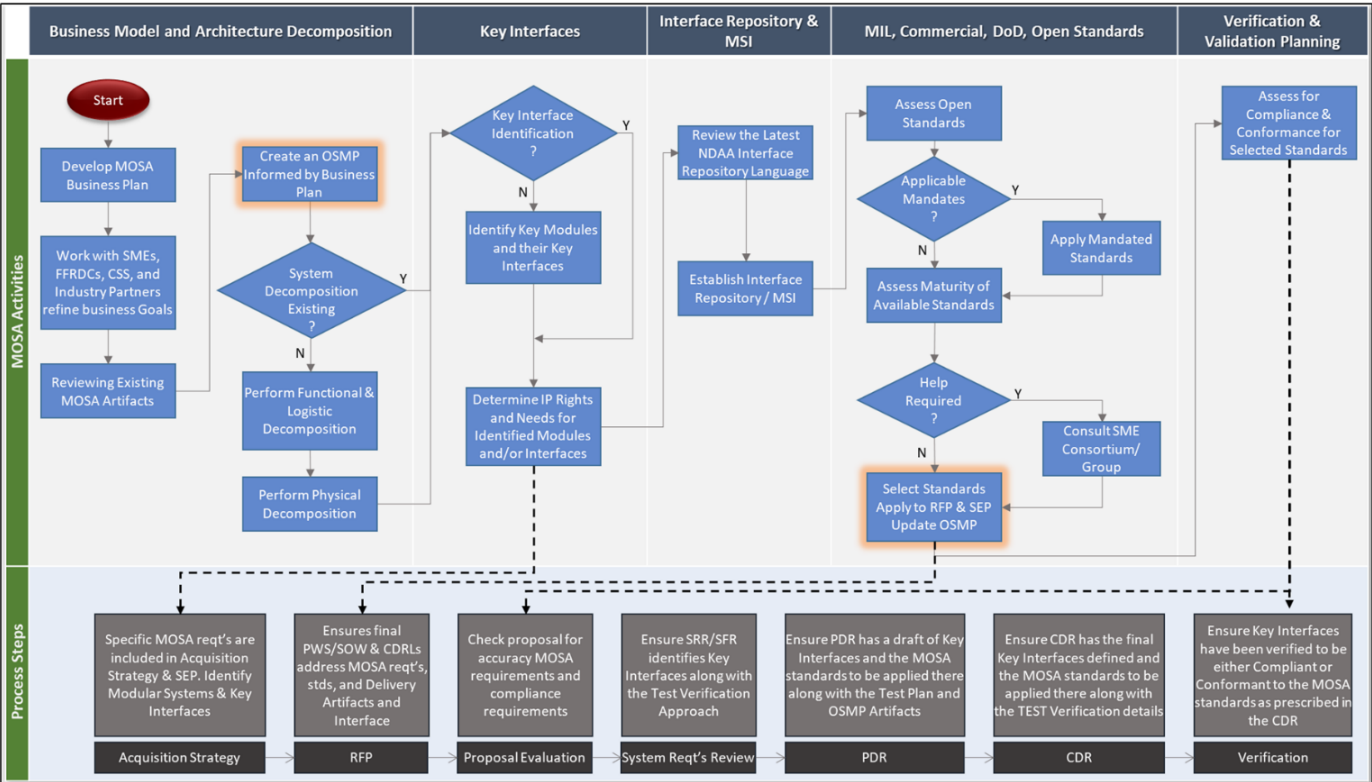


Figure 1 – A larger image located: Appendix B

NAVAIR’s framework offers a comprehensive, phased approach to implementing MOSA in system development. By aligning business strategies, technical decomposition, key interfaces, standards compliance, and verification, it ensures interoperability, modularity, and long-term adaptability of systems. The integration with acquisition and review processes ensures that MOSA principles are followed throughout the system lifecycle.

The image (Figure 2) visually represents three distinct approaches to system design—Extensive Modular Open Systems Approach, Selective Modular Open Systems Approach, and Proprietary/Non-Modular Design—each with varying degrees of modularity and interoperability. These approaches align with the Modular Open Systems Approach (MOSA) Process Framework, a structured methodology that guides system development and procurement through key phases such as (1) Business Model and Architecture Decomposition, (2) Key Interfaces, (3) Interface Repository & Modular Systems Interface (MSI), (4) Standards Compliance, and (5) Verification & Validation Testing.

1. Business Model and Architecture Decomposition: Systems are decomposed into modular, replaceable units, each with well-defined roles (e.g., sensor modules, processors, software services).

- Allows multiple vendors to compete on individual components.
- Enables technology insertion and upgrades without re-architecting the full system.
- Reduces lifecycle costs and supports competitive sustainment.

2. Key Interfaces: Key interfaces are the defined interaction points between modules—hardware, software, and data formats.

- Vendors build against open, government-owned or open standards-based interfaces.
- Promotes interoperability and module reusability.
- Encourages a marketplace of interchangeable components.

3. Modular System Interface (MSI) Repository: Central repository that stores interface definitions, version history, and usage guidance.

- Provides transparent access to key interface specs for all vendors.
- Reduces ambiguity and integration errors.
- Maintains consistency and traceability across programs and lifecycle phases.

4. Standards Compliance: Use of widely accepted, consensus-based technical standards (e.g., OMS, UCI, AMS-GRA, FACE®, SOSA®, WOSA, POSIX).

- Encourages multi-vendor participation.
- Reduces barriers to entry for new suppliers.
- Enables interchangeability and long-term supportability.

5. Verification & Validation Testing (VVT): Independent testing ensures that modules adhere to OSAs and open standards, meet requirements and interface correctly with the system.

- Reduces cost and schedule risk by catching issues early.
- Supports incremental certification of modules.
- Allows for pre-qualified vendor solutions, speeding up fielding.

MOSA is designed to enhance interoperability, modularity, and lifecycle adaptability, ensuring that system components can be upgraded, replaced, or integrated with emerging technologies without requiring complete system overhauls. The three approaches in the image illustrate different levels of adherence to MOSA principles and the associated trade-offs in system flexibility, cost, and performance.

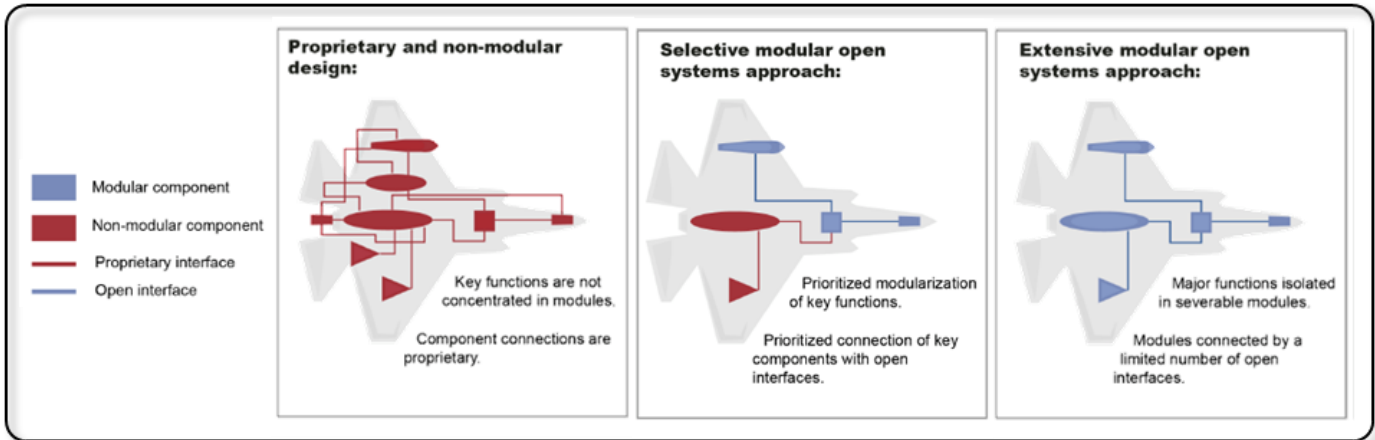


Figure 2

1. Proprietary and Non-Modular Design (Left Panel of the Image)

The leftmost section of Figure 2 illustrates a proprietary system architecture, where key functions are tightly integrated and dependent on closed, vendor-specific interfaces. Unlike the other two approaches, this design significantly limits interoperability and scalability, making upgrades and cross-platform integration challenging and costly. From a MOSA perspective, this approach is least aligned with modern system development best practices. The absence of open interfaces results in a system that faces vendor lock-in, increased lifecycle costs, and reduced flexibility in adapting to emerging technologies. Furthermore, compliance with open standards is minimal, contradicting MOSA's Standards Compliance phase, (which emphasizes adherence to military, commercial, and industry-wide interoperability requirements). The lack of modularity leads to complicated verification and validation efforts, increasing the risk of obsolescence and costly system-wide overhauls.

While proprietary designs may offer optimized short-term performance, they are increasingly being phased out in favor of modular, open architectures that support long-term technological evolution. **However, there are specific circumstances where a proprietary solution may be acceptable, even advantageous. For instance, a readily available Commercial Off-The-Shelf (COTS) product with a proprietary architecture could fulfill all program requirements "as is." This scenario is particularly relevant when facing urgent needs or time-sensitive deployments, where the speed of acquisition and integration outweighs the long-term benefits of a MOSA-compliant system. In such cases, accepting a proprietary solution can provide a rapid, cost-effective solution, especially if the required capability is well-defined and expected to have a limited lifespan or a clearly defined upgrade path is documented.**

Organizations transitioning from proprietary to modular approaches must carefully plan interface standardization and system decomposition strategies to ensure successful modernization. Even when initially selecting a proprietary system due to pressing needs, it's crucial to consider future interoperability, upgradability and develop potential migration strategies in order to minimize long-term vendor lock-in.

2. Selective Modular Open Systems Approach (Middle Panel of the Image)

The middle section of Figure 2 represents a targeted modular strategy, where selected system functions (those that are expected to be upgraded due to technology advancement, obsolescence, or may need modernization to pace an evolving threat) are modularized. This approach is evident in the selective use of modular components and open interfaces, balancing flexibility with cost constraints.

This approach aligns with MOSA's Key Interfaces phase, where specific interfaces are evaluated, defined, and integrated to support modularity without requiring complete system-wide redesign. The intellectual property (IP) implications of modularization must also be carefully managed, ensuring that proprietary constraints do not hinder future adaptability.

By prioritizing modernization efforts in key areas, this approach enables organizations to incrementally upgrade critical system components while maintaining compatibility with existing infrastructure. However, it requires careful standard selection and validation to avoid modularity inconsistencies. This is where Verification & Validation Planning in MOSA becomes essential, ensuring compliance with open standards and testing interoperability across modular components.

While less flexible than the Extensive Modular Open Systems Approach, this strategy is often more cost-effective and feasible, particularly for systems that require phased modernization rather than full-scale architectural transformation.

3. Extensive Modular Open Systems Approach (Right Panel of the Image)

The rightmost section of Figure 2 illustrates a fully modular system architecture, where major functions are isolated into self-contained modules (depicted in red and blue). These modules communicate through a comprehensive set of open interfaces, ensuring high interoperability and adaptability.

This approach aligns with the Business Model and Architecture Decomposition phase of MOSA, where organizations conduct system decomposition analysis (functional, logistical, and physical) and develop an Open Systems Management Plan (OSMP) to align modularity with strategic objectives. By maximizing the use of open interfaces, this approach facilitates seamless integration of new capabilities while mitigating risks associated with vendor lock-in and obsolescence.

Furthermore, the Interface Repository & Modular Systems Interface (MSI), established by DASN in 2025, plays a critical role in ensuring that these interfaces remain standardized across different system components. Despite its long-term benefits, implementing this approach requires rigorous governance and compliance with industry standards, as indicated in the Standards Compliance phase of MOSA. Organizations must evaluate mandatory and emerging standards to maintain system integrity and cross-platform compatibility.

In summary, (Figure 2) effectively illustrates the progression from rigid proprietary architectures to fully modular, open systems, demonstrating the trade-offs between interoperability, cost-efficiency, and long-term adaptability.

By following the MOSA Process Framework, organizations can ensure that their systems are future-proofed, cost-efficient, and aligned with evolving technological and industry standards. The integration of business strategies,

modular architecture design, key interface standardization, compliance assessment, and verification methodologies ensures that MOSA principles are embedded throughout the entire system lifecycle.

- The Extensive Modular Open Systems Approach represents the gold standard for MOSA implementation, offering maximum flexibility, interoperability, and long-term upgradeability.
- The Selective Modular Open Systems Approach provides a balanced strategy, enabling targeted modernization while maintaining some proprietary elements.
- The Proprietary and Non-Modular Design remains the least flexible, presenting challenges related to vendor dependence/lock-in, high lifecycle costs, and technological stagnation.

By leveraging MOSA principles and aligning system development with NAVAIR's structured framework, organizations can achieve modular, scalable, and efficient system architectures that meet the demands of evolving operational and technological landscapes.

Linking the MOSA Implementation Process to the OSMP DID

The MOSA Implementation Process is a structured framework designed to facilitate modularity, interoperability, and sustainability in system acquisitions. (Figure 3) The process begins with defining the **System and MOSA Objectives**, where stakeholder engagement plays a crucial role in aligning expectations and requirements. This phase establishes fundamental elements such as project objectives, quality attributes, architecture principles, and use cases that guide the system's development.

Once the foundational objectives are established, the process moves to the **Acquisition Plan**, which defines the MOSA objectives and selects the appropriate acquisition pathway. This phase also involves developing an intellectual property (IP) strategy to manage data rights¹ and accessibility, ensuring that system components remain flexible for future modifications. Additionally, contracting language is drafted to ensure compliance with MOSA principles, embedding modularity and open system requirements into legally binding agreements.

With the Acquisition Plan in place, attention shifts to the **Technical Plan**, which ensures the system's modularity and compliance with open system architectures (OSAs). This phase involves assessing appropriate modularity, evaluating applicable standards, and aligning system functionalities with the best-fitting OSAs. Identifying modular interfaces based on input/output needs and defining physical modularity requirements further ensures that the system can seamlessly integrate different components while maintaining adaptability.

¹ DoD OPEN SYSTEMS ARCHITECTURE June 2013, Contract Guidebook for Program Managers v.1

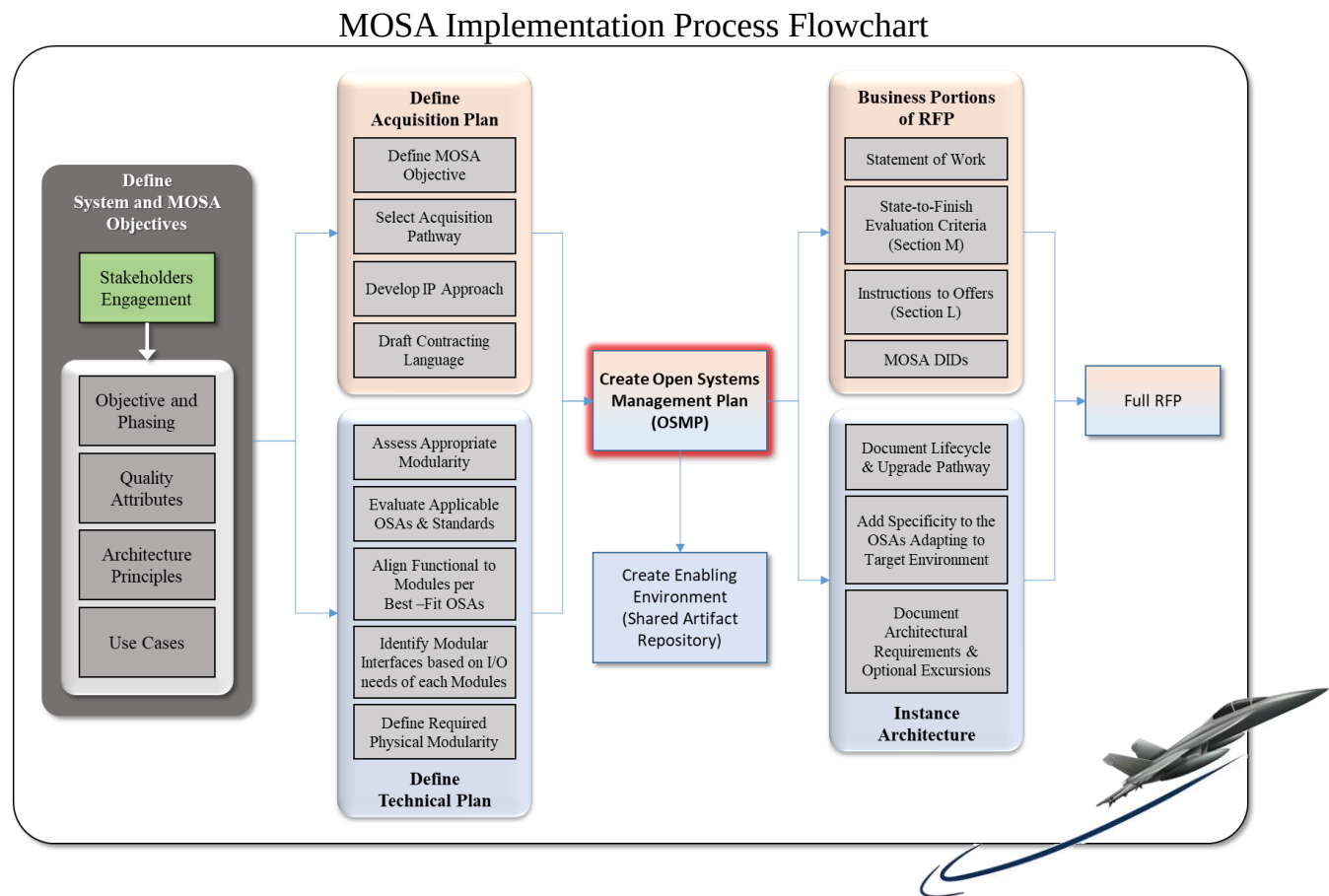


Figure 3

A critical component of this process is the **Open Systems Management Plan (OSMP)**, which serves as the guiding framework for implementing and managing the MOSA strategy throughout the system's lifecycle. This plan is complemented by the creation of an Enabling Environment, which provides a shared artifact repository containing technical documentation, system architectures, open standards, and reusable modules. This shared resource ensures consistency and interoperability across different projects.

The **Business Portions of the RFP (Request for Proposal)** formalize the acquisition strategy by defining the scope of work, evaluation criteria, proposal instructions, and Contract Data Requirements List (CDRLs) related to MOSA implementation. These business components are integrated into the full RFP to ensure that vendors and contractors adhere to MOSA requirements.

Finally, the **Objective Architecture** phase documents the system's lifecycle and upgrade pathways, ensuring that open system architectures are tailored to the target environment. This phase also specifies architectural requirements and optional excursions, providing the flexibility needed for future system adaptations.

By following this structured implementation process, organizations can ensure that their systems are designed with modularity, adaptability, and long-term sustainability in mind. This approach enables cost-effective system development while maintaining interoperability across defense and commercial sectors, ultimately supporting efficient and scalable technology evolution.

What is the Open System Management Plan (OSMP)

Open Systems Management Plan (OSMP) DI-MGMT-82099 Rev A – OSMP published in April 2022 superseding the original OSMP DID from January 2017. The essence of Open Systems Architecture (OSA) is an organized decomposition, using carefully defined execution boundaries, layered onto a framework of software and hardware shared services and a vibrant business model that facilitates competition. An OSMP describes the developer's approach for using modular design, standards-based interfaces, and widely-supported, consensus-based standards to achieve these goals. The OSMP shall include a developer's plans for a modular, open architecture, to include use of open interface standards and Reference Architecture (RA); design and management of, and dependencies between, Computer Software Configuration Items (CSCIs); design information documentation; technology insertion; life cycle sustainability; treatment of proprietary or vendor-unique elements; and reuse of existing items. The OSMP includes a support plan for the Modular Open Systems Approach (MOSA) and provides the acquirer visibility into the developer's open systems approach and implementation. This forms a basis for evaluation of a developer's proposal and adherence to MOSA technical and business goals.

The Open System Management Plan (OSMP) Partnership: A Procurement Strategy for NAVAIR Systems

The Open Systems Management Plan (OSMP) is not just a program document or contractual obligation – it's a strategic framework and living roadmap for ensuring the long-term viability and adaptability of NAVAIR systems. This document, grounded in the principles of openness and modularity, fosters a collaborative partnership between government and industry to guide the evolution of these systems throughout their entire lifecycle.

Government Leadership Sets the Vision:

The OSMP journey begins with strong government leadership. This goes beyond administrative oversight; it requires establishing a clear vision for the future of NAVAIR systems, encompassing:

- **Operational Outcomes:** Defining how NAVAIR systems will be used to counter future threats and meet how quickly the evolution of the mission needs.
- **System Capabilities:** Outlining specific functionalities and performance requirements necessary to achieve those outcomes.
- **Business Objectives:** Setting clear targets for cost, reuse, functional (modular) deposition, technology evolution, and performance throughout the system lifecycle.

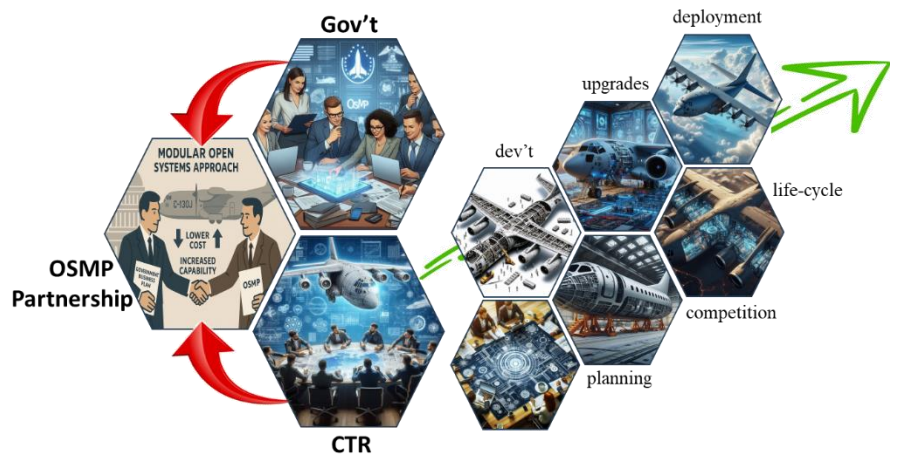
By embracing the Modular Open Systems Approach (MOSA) from the outset, the government lays the foundation for a flexible, interoperable, and resilient architecture. This top-down approach ensures alignment between technological decisions and strategic goals, such as reducing total ownership cost, increasing adaptability, and ensuring sustained operational relevance.

OSMP Partnership: A Collaborative Journey:

The OSMP thrives on a collaborative governance model, symbolized by the handshake between government and contractor (CTR) stakeholders. This partnership is not transactional but rather a relationship built on trust, mutual commitment, and shared success.

It's important to note that this collaboration is not a one-time event. The government initiates the OSMP process and continues to provide oversight throughout the project lifecycle. The contractor, in turn, expands upon the technical content of the OSMP, delivering regular updates at major milestone reviews or as needed by the program office. Here's a closer look at the respective roles:

- **Government's Role:** The government retains control of the strategic direction, defining the "why" and "what" of the NAVAIR system's evolution. It provides architectural oversight and ensures the contractor's work aligns with overarching objectives.
- **Contractor's Role:** The contractor provides technical expertise, translating the government's vision into reality. They deliver innovative, modular solutions that meet both platform-specific needs and broader strategic goals.



This dynamic partnership is characterized by a bi-directional flow of information and accountability. The government sets the vision, while the contractor is empowered to innovate within that framework, delivering cost-effective and upgradeable solutions.

From Development to Deployment: A Lifecycle of Continuous Evolution:

The OSMP is not a static document but rather a dynamic roadmap that guides NAVAIR systems through their entire lifecycle. This iterative process encompasses the major lifecycle milestones, but not limited to:

- **Planning:** Continuously assessing emerging threats, mission needs, and technological advancements to anticipate future requirements.
- **Development:** Designing and integrating new capabilities within a modular framework, enabling rapid technology refresh and competitive innovation.
- **Competition:** Leveraging modularity to foster competition for subsystems and components, driving down costs and stimulating innovation.
- **Lifecycle Management:** Implementing strategies to reduce sustainment burdens, mitigate obsolescence, and ensure long-term affordability.
- **Upgrades:** Pre-defined interface control documents (ICDs) and modular standards enable seamless integration of new technologies and capabilities.
- **Deployment:** Rapid and efficient fielding of upgraded systems with minimal disruption to the core platform.

This lifecycle approach, underpinned by digital engineering principles and a systems-of-systems architecture, ensures NAVAIR systems can adapt to future operational challenges and remain cornerstones of national security for decades to come.

The OSMP is more than just a plan – it’s a strategic investment in the future of NAVAIR systems. By fostering collaboration, embracing modularity, and adopting a lifecycle perspective, the OSMP ensures these critical assets remain adaptable, affordable, and operationally relevant for generations to come.

NAVAIR’s Open Systems Management Plan (OSMP) Process Framework

The Open Systems Management Plan (OSMP) process framework (Figure 4) provides a structured approach (with the appendixes referenced from the *OSMP DID*²) in developing an OSMP document, ensuring that modular open systems principles are effectively integrated into system development and lifecycle management. This process aligns with naval and program office business goals while delivering system increments through a Modular Open Systems Approach (MOSA). Below is a breakdown of the process framework:

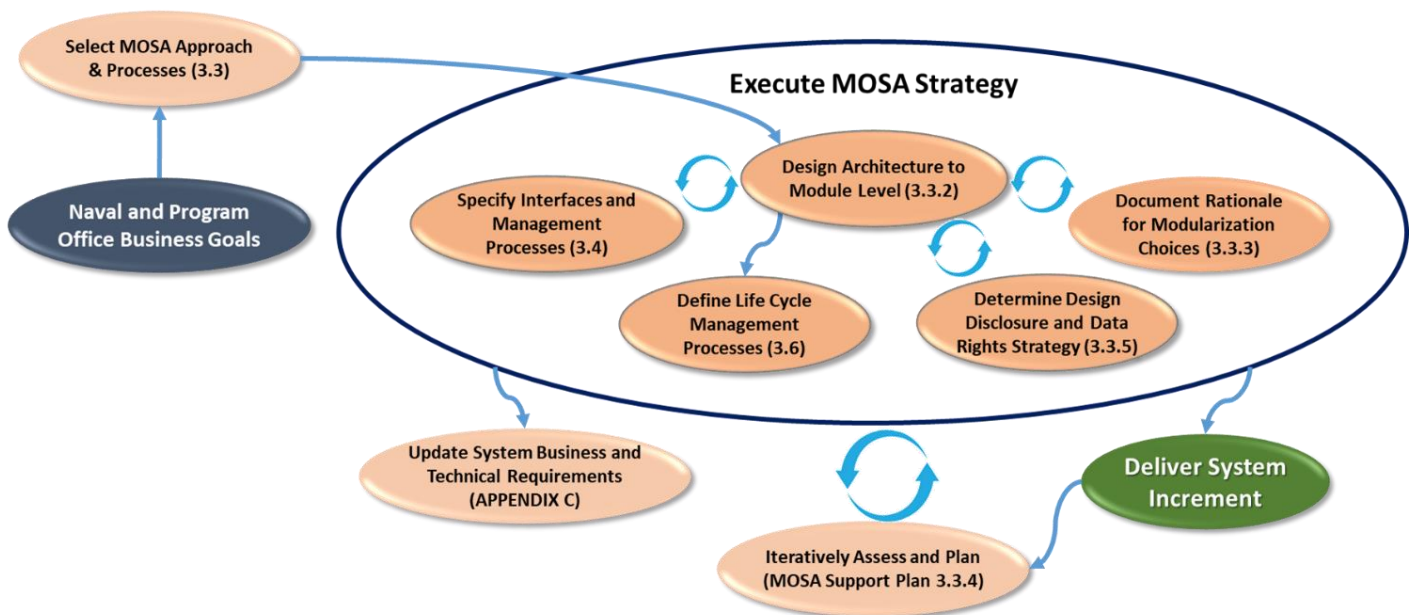


Figure 4

1. Aligning with Naval and Program Office Business Goals

At the foundation of the OSMP, process is the alignment with Naval and Program Office Business Goals. This ensures that the system development approach remains consistent with overarching organizational strategies and operational requirements.

² OSMP_DI-MGMT-82099A_2022.04.27

2. Selecting the MOSA & Processes Approach (3.3)

Once business goals are established, the next step is to select the MOSA & Processes Approach (3.3). This involves identifying the best methodologies for implementing a modular open system that achieves the business goals.

3. Executing the MOSA Strategy

The core of the OSMP process involves Executing the MOSA Strategy, which includes several key steps:

- **Design Architecture to Module Level (3.3.2)**
A critical aspect of MOSA execution is breaking down the system into modules with clearly defined functionality and open interfaces, enabling easier upgrades and modifications.
- **Document Rationale for Modularization Choices (3.3.3)**
Justification for modular decisions must be documented, outlining why specific modularization strategies were chosen and their expected benefits.
- **Determine Design Disclosure and Data Rights Strategy (3.3.5)**
This step addresses intellectual property considerations, ensuring that data rights, proprietary components, and open-source elements are appropriately managed.
- **Specify Interfaces and Management Processes (3.4)**
This step defines how various system components interact and are managed. It ensures that standardized interfaces are in place to support modularity.
- **Define Lifecycle Management Processes (3.6)**
Life cycle management is crucial to maintaining and upgrading the system over time. This step defines policies and strategies for sustaining modular components.

4. Updating System Business and Technical Requirements

As part of iterative refinement, it is essential to update System Business and Technical Requirements (Appendix C) to reflect evolving needs and insights gained during MOSA execution.

5. Iterative Assessment and Planning (MOSA Support Plan 3.3.4)

To ensure continuous improvement and alignment with system goals, the process includes an iterative cycle of Assessing and Planning (MOSA Support Plan 3.3.4). This allows for adaptive adjustments as the system matures.

6. Delivering System Increments

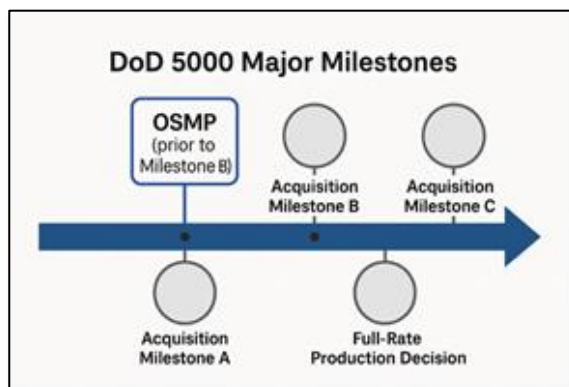
Finally, the process culminates in the Delivery of System Increments, ensuring that modular components are effectively integrated and delivered in stages to maximize system efficiency and responsiveness.

The OSMP process framework ensures a systematic approach to implementing MOSA principles, allowing for flexibility, interoperability, and sustainment of system capabilities over time. By following this structured methodology, organizations/Program Offices can efficiently develop, manage, and evolve their systems while maintaining alignment with business goals and technological advancements.

Strategic Alignment of the OSMP and the DoD Acquisition 5000 Framework

The successful development and implementation of an Open System Management Plan (OSMP) within a Department of Defense (DoD) acquisition program necessitates a strategic and proactive alignment with the DoD Acquisition 5000 framework from the earliest possible stages of the acquisition lifecycle. This alignment is not merely a procedural requirement but a fundamental element in fostering adaptable, cost-efficient, and ultimately more effective defense capabilities. By integrating open systems principles – encompassing modularity, interoperability, and open architectures – at program inception, acquisition professionals can proactively shape system requirements, inform technology selection, and strategically formulate the acquisition approach to mitigate long-term risks and optimize lifecycle costs.

The DoD Acquisition 5000 framework, with its phased approach and milestone decision points, provides a structured environment for this alignment. The OSMP is strategically integrated and refined across these key phases:



Pre-Milestone A: Foundational Integration during Materiel Solution Analysis

The Materiel Solution Analysis phase represents the optimal juncture to embed open systems thinking into the program's 'DNA'. At this stage, the focus is on identifying potential solutions to validated capability gaps. Integrating open systems principles early facilitates:

- **Informed Technology Selection:** Prioritizing the evaluation and selection of technologies that inherently support modular, scalable, and interoperable architectures. This early consideration can steer the program away from proprietary solutions that might impede future upgrades or integration efforts.
- **Proactive Requirements Development:** Embedding open systems considerations directly into the initial requirements definition ensures that these principles are not afterthoughts but foundational design tenets. This includes specifying interfaces, data standards, data rights, modularity requirements, and conformance verification tests from the outset.
- **Strategic Acquisition Planning:** Formulating an acquisition strategy that actively mitigates long-term risks associated with vendor lock-in and cost inefficiencies. This involves exploring contracting mechanisms that incentivize the adoption of open, standards-based solutions and promote competition.
- **Conceptualizing the OSMP:** Initiating the development of the OSMP as a guiding document that aligns with the overarching program objectives and the principles of the DoD 5000 framework. This early conceptualization ensures that open systems considerations are central to the program's strategic direction.

Milestone A: Formalizing the OSMP within the Acquisition Strategy

Upon reaching Milestone A, which signifies the approval to enter the Technology Maturation and Risk Reduction phase, the OSMP is formalized and explicitly integrated into key acquisition planning documents, most notably the Acquisition Strategy. This formal integration ensures that:

- **The Materiel Solution Embraces Open Architecture:** The selected materiel solution is demonstrably aligned with program-specific open architectures and standards, providing a foundation for future flexibility and interoperability.
- **The Acquisition Framework Supports Open Systems:** The acquisition framework, including contracting approaches and evaluation criteria, explicitly incorporates open systems principles into procurement and development activities.
- **The OSMP Serves as a Guiding Framework:** The formalized OSMP acts as a central reference point for subsequent program phases, ensuring consistent alignment with the DoD 5000 framework and the program's open systems goals.

Milestone B: Refining the OSMP for Technology Maturation and Risk Reduction

As the program transitions into the Technology Maturation and Risk Reduction phase at Milestone B, the OSMP requires further refinement to effectively guide:

- **Technology Development Alignment:** Ensuring that technological development efforts are explicitly aligned with established open architecture frameworks and standards. This includes the selection of system/subsystem modulization with appropriate interface standards, data formats, and communication protocols.
- **Risk Mitigation Strategies:** Developing and implementing risk reduction strategies that specifically address the challenges associated with proprietary solutions and vendor lock-in. This might involve prototyping efforts focused on demonstrating the viability of open interfaces and modular designs.
- **Adaptable Technology Selection:** Prioritizing the selection of adaptable technologies that inherently promote long-term system flexibility, maintainability, and interoperability. This includes considering the potential for future upgrades, technology insertions, and integration with other systems.

Sustained Alignment Across the Entire Acquisition Lifecycle

The OSMP is not a static document to be filed away after Milestone B. Its relevance and effectiveness depend on its continuous evolution and application throughout the entire acquisition lifecycle. Key phases where the OSMP is revisited and updated include:

- **Engineering and Manufacturing Development (Milestone C):** Ensuring that the detailed design and implementation of the system remain consistent with the principles of modularity and extensibility outlined in the OSMP.
- **Production and Deployment:** Facilitating the seamless integration of open systems solutions into operational capabilities, ensuring that the deployed systems adhere to the established open architecture standards.
- **Operations and Support:** Enabling cost-effective system sustainment, modernization, and technology refresh cycles by leveraging the inherent flexibility and interoperability of an open systems approach.

The Strategic Imperative for Early OSMP Alignment

The benefits of aligning the OSMP early with the DoD 5000 framework are multifaceted and strategically significant:

1. **Shaping System Requirements:** Early integration of open systems principles ensures that modularity, interoperability, and technology reuse are not merely desirable attributes but, instead, are fundamental requirements driving system design and procurement strategies.
2. **Optimizing Acquisition Strategy:** The OSMP serves as a critical enabler for developing contracting approaches that foster vendor competition, incentivize the adoption of open, standard-based solutions, and ultimately lead to more innovative and cost-effective acquisitions.
3. **Mitigating Cost and Risk Factors:** By proactively reducing reliance on proprietary technologies and fostering a competitive vendor environment, an open systems approach significantly enhances cost-effectiveness, lifecycle sustainability, innovation, and the overall agility of the acquisition program.
4. **Enhancing Interoperability and Future Integration:** An early and sustained emphasis on open architecture ensures that newly acquired systems can seamlessly integrate with existing and future defense technologies, thereby bolstering overall operational effectiveness and mission readiness.



In summary, the development of an effective OSMP for a DoD acquisition program demands a strategic and proactive alignment with the DoD 5000 framework, commencing at the beginning of the Materiel Solution Analysis phase (Pre-Milestone A and Milestone A) and maintained as a dynamic and evolving document throughout the entire acquisition lifecycle. This early and continuous alignment is not simply a best practice; it is a strategic imperative for ensuring the long-term adaptability, cost efficiency, and interoperability of defense systems in an ever-evolving technological landscape. By embracing open systems principles from the outset, DoD acquisition professionals can significantly enhance the likelihood of delivering effective and sustainable warfighting capabilities.

Proposal Evaluation and Contract Award Guide

A structured approach to evaluating and comparing offerors' open systems strategies involves requiring a draft OSMP as part of the proposal submission. This draft, which is derived from the government's initial baseline to the contractor's; should address key areas, including scope, open architecture approach, data rights ³strategy, and interface management. The following Request for Proposal (RFP) Section L language is recommended for tailoring and inclusion in solicitations:

“The Offeror SHALL prepare and submit a draft OSMP in accordance with (IAW) OSMP DID [DI-MGMT-82099 Rev A], detailing the development of an open system. This includes their open architecture approach IAW OSMP DID Section 3.3.1, design disclosure and data rights strategy IAW OSMP DID Section 3.3.5, and interface design and management IAW OSMP DID Section 3.4.”



To ensure a comprehensive evaluation, the following RFP Section M: Evaluation Factors for Award language is also recommended for inclusion:

“The Government will evaluate the extent to which the proposed approach supports the development and sustainment of an open system. This includes considerations such as design disclosure, data rights strategy, interface design, and interface management, assessing whether the approach is thorough, complete, adequate, feasible, and adaptable.”

“The Government will assess the offeror's approach to managing sub-vendor relationships, ensuring that expectations for openness and modularity are effectively flowed down.”

Additionally, to establish a structured approval process, an initial OSMP submission is required shortly after contract award (CA). This initial delivery should, at a minimum, address the same key sections (3.3.1, 3.3.5, and 3.4) specified in the proposal phase. Given the early stage of development, some sections, such as architecture and modularity decisions, may be incomplete or contain only preliminary information.

Recommended contract language

The contractor shall develop and maintain an OSMP throughout the project lifecycle, as detailed in OSMP. This requirement will be clearly stated within the Statement of Work (SOW), referencing the governing DID to establish its contractual importance.

“The Contractor SHALL deliver the initial OSMP in accordance with (IAW) OSMP DID [DI-MGMT-82099 Rev A], providing details on the development of an open system, including their open architecture approach IAW OSMP DID Section 3.3.1, design disclosure and data rights strategy IAW OSMP DID Section 3.3.5, and interface design and management IAW OSMP DID Section 3.4.”

³ For more on Data Rights; reference DoD OPEN SYSTEMS ARCHITECTURE June 2013, Contract Guidebook for Program Managers v.1

Milestones

The OSMP is a living document, evolving throughout the program lifecycle to reflect the latest architectural decisions, including modularity, interface definitions, and data rights considerations. To maintain alignment with program objectives, the contractor shall deliver updated OSMPs prior to each major program milestone. Each update will include the complete OSMP document, incorporating the latest architectural decisions, design modifications, and relevant program changes, along with a detailed summary of key changes from the previous iteration, highlighting modified areas and their rationale. The CDRL will specifically reflect this iterative process by having recurring entries for updated OSMP deliverables.

It is important to establish a well-defined process for joint government-contractor review and approval of OSMP updates to encourage ongoing communication and alignment. This collaborative approach, combined with the milestone-driven delivery schedule, guarantees the OSMP remains a relevant and valuable tool for communication, informed decision-making, and successful open system modernization.

Contracting for MOSA Using an OSMP

"The Contractor SHALL develop, maintain, and deliver an Open Systems Management Plan (OSMP) in accordance with OSMP DID [DI-MGMT-82099 Rev A]."

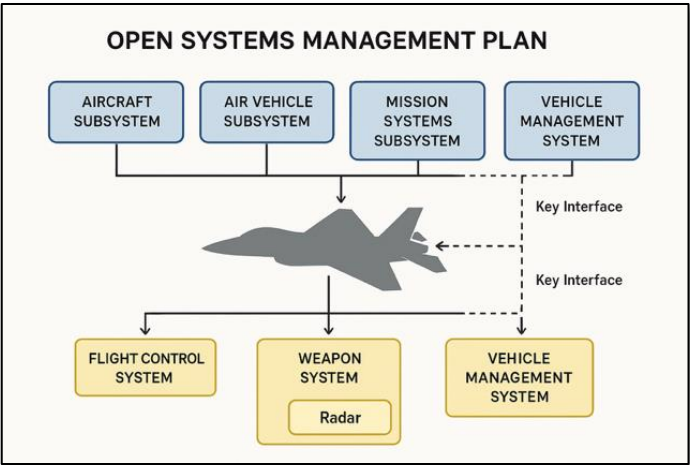
"The Contractor SHALL provide a detailed summary of changes between the latest OSMP and the previously delivered version, in compliance with the Technical Report-Study/Services DID [DI-MISC-80508 Rev B]."

Systematic tracking, review, and approval of OSMP updates will enhance transparency, facilitate strategic decision-making, and drive affordability and innovation throughout the program lifecycle.

Key Business and Technical Drivers to Consider in an OSMP

The Open System Management Plan (OSMP) is a critical framework for ensuring system modularity, interoperability, scalability, and cost-effectiveness throughout its lifecycle. It achieves this by enforcing open architectures, standardized interfaces, and modular system design (including module function description) to prevent vendor lock-in, enable competitive sustainment strategies, and support rapid technology insertion.

Successful OSMP implementation requires a business-driven functional decomposition, aligning technical decisions with cost, competition, scalability, and long-term sustainment strategies to avoid unnecessary complexity and costs.



To ensure contractor compliance with the OSMP, the Program Office requires a system-level test plan which incorporates a Verification and Validation Testing (VVT) framework. This framework verifies compliance with the OSMP's architectural design and validates the jointly developed acquisition and sustainment strategies. This ensures the system adheres to the Modular Open Systems Approach (MOSA) outlined in the OSMP throughout its lifecycle. This ensures that:

1. The contractor delivers all required artifacts as per the OSMP.
2. The system aligns with open architecture principles and business objectives.
3. The final system implementation meets functional, key interfaces/standards conformance, security, and interoperability requirements.
4. The government has a structured contractor acceptance process to validate all deliverables before final approval.

By enforcing strict acceptance criteria, the government ensures that the final system is open, modular, interoperable, cost-effective, and future-proof, preventing vendor lock-in and enabling strategic sustainment. This rigorous approach, combining business drivers with technical specifications, forms the foundation of a robust and adaptable system acquisition strategy.

To further detail the implementation and oversight of the OSMP, the following sections delve into the technical specifications, architectural components, and the critical verification and validation processes.

Technical Description of OSMP

1. Open System Principles

The OSMP enforces the following core principles:

- **Modularity:** System components must be independently upgradable and replaceable to prevent proprietary lock-in.
- **Interoperability:** Systems must follow open standards (e.g., SOSA, OMS/UCI, XML, etc) to ensure multi-vendor integration.
- **Scalability:** The architecture should support future expansion without significant re-engineering costs.
- **Sustainability:** Long-term maintenance and technology refresh should be cost-effective and vendor-agnostic.
- **Security & Compliance:** The system must adhere to Zero Trust principles, NIST 800-171, and DoD cybersecurity frameworks.

2. Architectural Components of OSMP

Component	Description
Hardware Abstraction Layer (HAL)	Decouples software from specific hardware implementations, ensuring longevity and flexibility.
Open Interfaces & APIs	Standardized APIs (e.g., REST, DDS) ensure interoperability with third-party components.
Middleware & Data Exchange	Implemented using DDS, MQTT, and service-oriented architectures (SOA) for real-time data distribution.
Security Framework	Zero Trust security model with multi-layer authentication, encryption, and real-time monitoring.
Software Modularity	Microservices-based design using containerization (Docker, Kubernetes) to ensure platform independence.
Continuous Integration/Continuous Deployment (CI/CD)	Automated testing and deployment pipelines to support agile development.

3. Business-Driven Functional Decomposition

3.1.1 Rationale for Business-Driven Functional Decomposition

An open system is not valuable unless it aligns with business goals such as cost reduction, competitive sustainment, scalability, and adaptability to emerging technologies. BDFD ensures that system decomposition aligns with economic and strategic imperatives rather than purely technical considerations.

3.2. Functional Decomposition Mapped to Business Objectives

Business Objective	Technical Functional Requirement	OSMP Contribution
Reduce Lifecycle Costs	Modular design for component replacement	Open standards allow competitive sustainment
Enhance Competition	Vendor-agnostic software interfaces	Standardized APIs and middleware
Support Rapid Innovation	Plug-and-play hardware/software modules	Compliance with MOSA, OMS, WOSA, and SOSA
Increase Security Resilience	Zero Trust architecture with continuous monitoring	Secure-by-design implementation
Enable Scalability & Technology Insertion	Dynamic software deployment model	CI/CD, containerized microservices
Prevent Vendor Lock-in	Full government ownership of technical baselines	TDP compliance, open IP rights

4. Verification & Validation Testing (VVT) Plan

4.1. Government Oversight in Verification

The Program Office must ensure that:

- 1. All contractor deliverables conform to OSMP specifications.
- 2. The system meets MOSA, Open Architecture (OA), key interfaces/standards compliance, security, and lifecycle sustainment requirements.
- 3. Independent Government testing and/or artifacts to verify contractor claims.

4.1.1. Verification Levels

Verification Level	Government Action
Artifact Verification	Audits of architecture documents, interface control documents, TDPs
Design Compliance	Hardware & software design reviews, modularity assessment
Implementation Validation	Source code audits, configuration management verification
Independent Testing	IV&V execution, third-party assessments

4.2. Contractor Acceptance Process

The government must formally accept or reject contractor deliverables based on:

- 1. Alignment with Business Objectives: Does the system meet the cost, competition, and sustainment goals outlined in the open business goals within the OSMP
- 2. Technical Compliance: Does the system conform to the open standards and technical requirements outlined in the OSMP?
- 3. Testing Results: Have all functional, security, and interoperability tests been passed successfully?

4.2.1. Contractor Acceptance Workflow

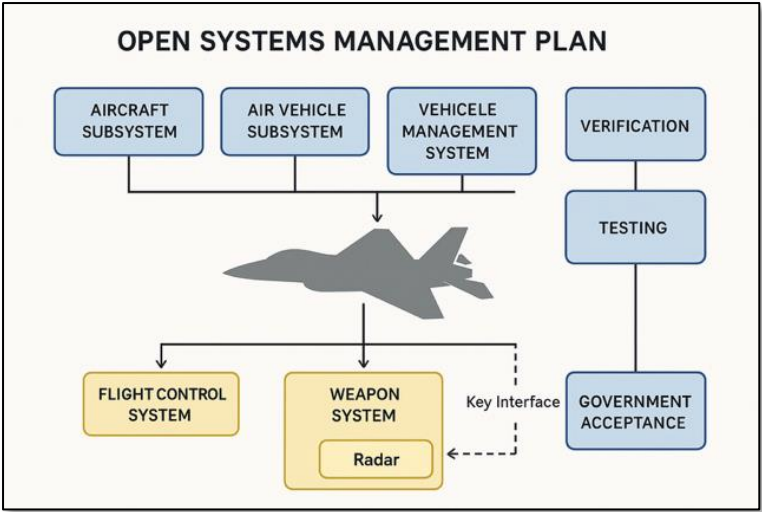
Phase	Government Action	Acceptance Criteria
Initial Review	Evaluate technical and business documentation	Meets BDFD & OSMP criteria
Artifact Validation	Verify architecture, interface control, and software documentation	Compliance with open standards
Test Execution	Conduct functional, interoperability, performance, and security tests	Meets or exceeds required thresholds
Independent Government Testing	IV&V assessment for unbiased evaluation	Confirms contractor results
Final Review & Approval	Assess cost/sustainment strategy and final compliance	Meets business and technical goals

4.3. Testing Methodology

The OSMP Contractor Acceptance Plan ensures that government oversight is comprehensive and structured, covering:

- Technical verification
- Business alignment
- Independent validation

By enforcing strict acceptance criteria, the government ensures that the final system is modular, interoperable, cost-effective, and future-proof, preventing vendor lock-in and enabling strategic sustainment.



Basic testing could incorporate the following, however but may not be limited to: four (4) types of tests: **functional testing** to ensure it operates correctly, **interoperability testing** to confirm compatibility with multiple vendors, **performance testing** to evaluate its behavior under stress, and **security testing** to find and fix vulnerabilities. Various tools are listed for each test type. For **final acceptance**, the contractor's work must meet the OSMP and business goals, government tests must confirm all requirements, and a practical, affordable sustainment plan must be in place.

By aligning business objectives with technical specifications through a business-driven functional decomposition and implementing a robust Verification and Validation Testing (VVT) framework, the government can ensure contractor compliance, prevent vendor lock-in, and foster competitive sustainment strategies. This comprehensive approach, encompassing open architecture principles, standardized interfaces, and modular design, ultimately enables the acquisition of adaptable, sustainable, and secure systems capable of meeting current and future needs.

Concluding Thoughts

In conclusion, navigating the complexities of modern naval warfare demands a paradigm shift in how we equip our forces. The Modular Open Systems Approach (MOSA) is not merely a desirable option; it is an essential strategy for achieving a more agile, adaptable, and fiscally responsible future for naval aviation. By resolutely embracing open architectures and modularity, we unlock a powerful engine for rapid technological advancement, seamless multi-vendor interoperability, and sustained competition – vital elements for maintaining our strategic advantage.

The absolute linchpin of successful MOSA implementation is a meticulously crafted and aggressively managed Open Systems Management Plan (OSMP). This is far more than a document; it is the living blueprint, the dynamic compass guiding every stage of the system lifecycle. Critically, the initial development of the OSMP by the government must be laser-focused on clearly articulating their core business goals and non-negotiable technical needs. This foundational "first cut" by the government is paramount. It sets the strategic direction, ensuring that all subsequent efforts are rooted in achieving tangible outcomes for the warfighter and responsible resource management for the taxpayer.



It is at this crucial juncture that the contractor's expertise becomes invaluable. They will then build upon this robust government foundation, layering in their innovative solutions and technological capabilities to directly address the pre-defined government objectives. This collaborative process, guided by the OSMP, fosters a synergistic environment where government vision and industry ingenuity converge to create a unified solution. Ultimately the desired outcome is a system that not only meets the warfighter's evolving operational demands with cutting-edge technology but also does so in a manner that demonstrates responsible stewardship of taxpayer dollars.

Therefore, a deep and unwavering commitment to MOSA principles, manifested through a comprehensive, government-led "first cut" OSMP that clearly defines business goals and technical needs, is not just beneficial – it is mission-critical. This proactive and strategic approach empowers programs to deliver the flexible, adaptable, and sustainable systems necessary to maintain our decisive edge in an increasingly complex operational environment. The OSMP, as a continuously evolving framework, remains the unwavering bedrock for supporting our warfighters and ensuring a future of technological superiority and fiscal responsibility in naval aviation.

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Referenced Documents

Assistant Secretary of the Navy (ASN) – Naval Modular Open System Approach – Guidebook v1.0. Published by ASN(DASN), December 2024

10USC Title 10, United States Code (U.S.C.)

MOSAACQ Office of the Secretary of Defense (OSD) Program Manager's Guide: A MOSA to Acquisition, v2.0, published by OSD, September 2004

OSMPA Open Systems Management Plan (OSMP) Data Item Description (DID) Rev A [DI-MGMT82099A], published by the Department of Defense (DoD) Defense Standardization Program Office, April 27, 2022

2023 Contracting for MOSA Using an OSMP whitepaper, published by Army, September 2023.

Figure 2; Source: GAO analysis of Department of Defense Information. | GAO-25-106931

DoD OPEN SYSTEMS ARCHITECTURE June 2013, Contract Guidebook for Program Managers v.1

Appendix A:

Common Mistakes to Avoid in Implementing MOSA

A common challenge in discussions about the Modular Open Systems Approach (MOSA) is the widespread, yet often incomplete, understanding of its principles and applications. Many professionals recognize that MOSA involves modularity and open interfaces, but their comprehension frequently does not extend beyond these basic concepts. This gap in understanding can lead to inconsistent implementations and miscommunications. Therefore, when discussing or describing MOSA, it is crucial to explicitly state and cite its definition and principles to ensure a shared and accurate understanding across all stakeholders.



Defining MOSA: A Legal and Strategic Framework

MOSA is legally defined under 10 U.S.C. 4401 as an integrated business and technical strategy that:

- Employs a modular design that leverages well-defined modular system design and open interfaces.
- Undergoes verification to ensure that modular system interfaces conform to established standards.
- Utilizes a system architecture that enables severability, allowing components to be upgraded or replaced without disrupting the entire system.
- Complies with technical data rights guidance as outlined in 10 U.S.C. 3771 through 3775, ensuring appropriate management of intellectual property and proprietary elements.

Beyond its legal definition, MOSA is also guided by best practices documented in “The OSD Program Manager’s Guide”: A MOSA to Acquisition, v2.0 (September 2004) [MOSAACQ], which identifies five fundamental principles essential to its successful implementation:

The Original Five Core Principles of MOSA

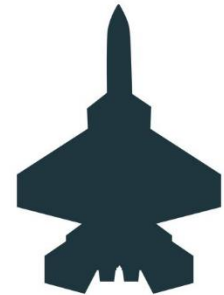
1. **Establish an Enabling Environment** – Programs must create policies, governance structures, and business models that support modularity and openness.
2. **Employ Modular Design** – System architecture is decomposable into discrete, interchangeable modules that enhance flexibility and scalability.
3. **Designate Key Interfaces** – Open interfaces are explicitly defined to ensure interoperability and future upgrades without vendor lock-in.
4. **Use Open Standards** – Standardized, publicly available specifications are used to maximize cross-platform compatibility and industry adoption.

5. **Certify Conformance** – A structured verification process must be in place to ensure compliance with modularity, interface, and standardization requirements.

MOSA vs. OSA: Understanding the Distinction

A frequent misconception about MOSA is the assumption that it refers to a specific system architecture. However, MOSA is not an architecture—it is an approach. The "A" in MOSA stands for "Approach," not "Architecture." This distinction is critical because MOSA outlines a strategic methodology for achieving an open, modular system rather than prescribing a fixed architectural framework.

By contrast, Open Systems Architecture (OSA) represents the tangible architectural implementation that results from applying MOSA principles. Programs do not simply "adopt" an OSA; rather, they must establish and adhere to a MOSA-based methodology to develop a system architecture that embodies openness, modularity, and interoperability.



Key Components of an Open Systems Architecture (OSA)

For an OSA to be valid and effective, it must incorporate three essential elements:

1. **Modularity** – The system must be structured as independent, loosely coupled modules that can be developed, tested, and upgraded separately.
2. **Open Interfaces Using Open Standards** – Interfaces are defined according to widely accepted, publicly available standards to ensure interoperability across different vendors and technologies.
3. **Verification Mechanism** – A formalized process must be in place to verify both modularity and interface openness to ensure compliance and prevent unintended integration issues.

If any aspect of a MOSA-based implementation does not **directly** contribute to achieving one or more of these core architectural elements, the approach must be re-evaluated and refined.

Avoiding Common MOSA Misuses and Buzzwords

The increasing emphasis on MOSA in government and industry has led to its frequent misuse as a **buzzword**, often applied incorrectly or imprecisely. To maintain clarity and fidelity to its intended principles, the following common mistakes should be avoided:

- **"MOSA interfaces"** → The correct term is **"open interfaces."** MOSA itself is not an interface type but rather an approach that promotes the use of open interfaces.
- **"MOSA standards"** → The correct term is **"open standards."** Standards are not exclusive to MOSA but are a fundamental aspect of ensuring interoperability.
- **"MOSA requirements"** → Instead of using this vague phrase, requirements should be specified more precisely, such as:

- **Modularity requirements** (defining system decomposition and integration points)
- **Open interface requirements** (ensuring interfaces conform to non-proprietary, open standards)
- **Conformance requirements** (establishing validation and verification criteria)

As part of adhering to or adopting an Open Standard, it is critical to understand the difference between conformance vs compliance. An Open Standard may specify conformance requirements whereas another competing Open Standard may only require compliance. It is critical for the Government to understand the difference between Compliance vs Conformance, and how each approach satisfies the intent highlighted in the business goals highlighted within the OSMP. In short, conformance to programs MOSA should not be conflated with conformance to open standards i.e. certain standard conformance approaches may not be cost effective and simply validating compliance by assessing a system or its components modularity and adherence to key open interfaces may suffice.

Appendix B:

NAVAIR's Modular Open Systems Approach (MOSA) Process Framework is on the next page.



